

Lecture 0

Why integrability?

Objectives. *Two experiments in which a system of many interacting particles refused to forget how it was started. One line of mechanics that says why one dimension is special. What integrability means, classically and quantum mechanically, in words and pictures, before any definition. The single consistency condition — three lines crossing — behind the commuting transfer matrices that carry Part I into Part II. And a map: where the twenty-three lectures go, and where they deliberately do not. Nothing here is used later as a premise. Each technical claim returns in the lecture it names, where it is proved, sketched or cited, and marked as such.*

0.1 Two experiments that should not have happened

The Fermi–Pasta–Ulam–Tsingou chain

Take N equal masses in a row, joined by springs, with the two ends fixed. If the springs obey Hooke's law the problem is linear and completely solved: the motion is a superposition of N normal modes, the k -th a standing sine wave with frequency

$$\omega_k = 2 \sin \frac{k\pi}{2(N+1)}, \quad k = 1, \dots, N \quad (0.1)$$

(in units with mass and spring constant one). Each mode is a harmonic oscillator on its own. The energy in each mode is conserved separately: N conserved quantities, one per mode. Start the chain in the lowest mode and it stays there for ever.

Now make the springs slightly nonlinear. With r the extension of a spring, take the spring energy

$$V(r) = \frac{r^2}{2} + \alpha \frac{r^3}{3}, \quad (0.2)$$

with α small. The modes are now coupled: the cubic term lets energy flow from one to another. What should happen seemed clear to Fermi (Toda, §1.1). The expectation, from statistical mechanics, was that a large system of coupled degrees of freedom, left alone, forgets everything about how it was started except its conserved quantities — here, only the total energy — and settles into thermal equilibrium, where on average every mode holds the same share of the energy. That is *equipartition*. The nonlinearity was put in precisely to let it happen, and the plan was to watch how long it took.

Fermi, Pasta and Ulam, with Mary Tsingou writing and running the programs on one of the first electronic computers at Los Alamos, integrated this chain with 32 and with 64 masses, all the energy in the lowest mode at the start (Toda, §1.1). The 1955 report thanked Tsingou for coding and running the computations; her decisive part was properly recognised, and her name added to the problem's, only about fifty years later (Dauxois 2008). [Figure 1](#) is the same experiment, integrated for these notes.

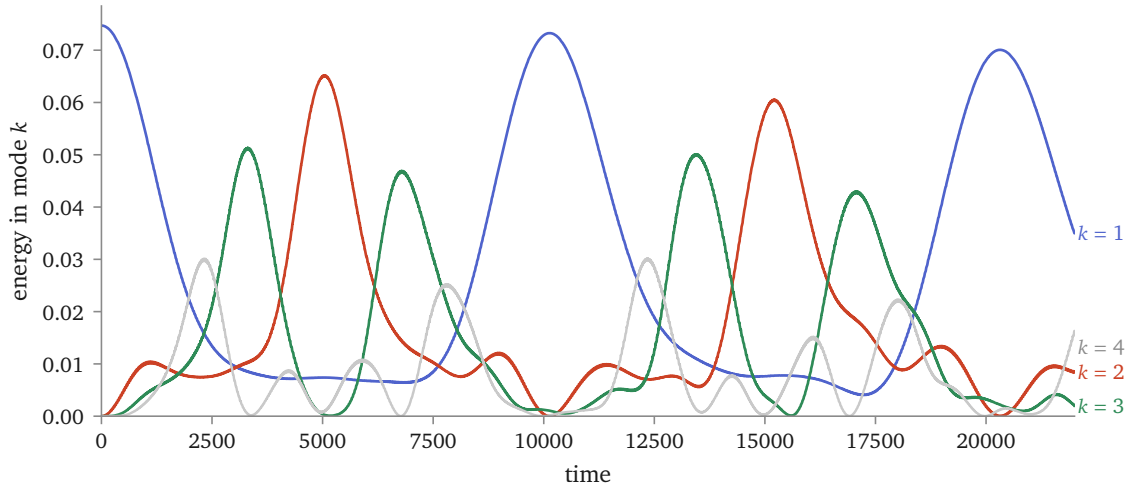


Figure 1: The Fermi–Pasta–Ulam–Tsingou recurrence, integrated here rather than quoted: 32 masses, $\alpha = \frac{1}{4}$, ends fixed, started at rest with all the energy in the lowest mode. Plotted is the energy in each of the four lowest modes. The energy leaves mode 1 (below 10% of its start), passes through modes 2, 3 and 4 — and comes back: more than 95% of it is in mode 1 again near $t = 10^4$, about 150 periods of that mode, and more than 90% a second time near $t = 2 \times 10^4$. Modes 9 to 32 together never hold even 1% of the energy; equipartition would give them three quarters of it (`scripts/figures/fput_recurrence.py`, which asserts each of these numbers).

The energy did spread — into the next few modes, never into the rest — and then it came back. Not equipartition: a *recurrence*, the chain returning almost exactly to where it started, again and again.

It is not an artefact of the integration: in Figure 1 the total energy is conserved to a few parts in a million over the whole run, and the recurrence has been reproduced many times since (Toda, §1.1). What it says is that the chain behaves as though it had *more conserved quantities than energy*. Energy alone cannot stop the flow into the high modes: every way of sharing out the same energy is allowed by that one constraint. Something else is holding the motion back. The Hooke chain had N conserved quantities and never forgot anything; the question FPUT raised is how much of that survives the nonlinearity.

The quantum Newton’s cradle

Fifty years later the same question was asked of a quantum system, with atoms instead of a computer. Kinoshita, Wenger and Weiss (2006) trapped rubidium-87 atoms in an array of thin tubes made of laser light, each tube holding between 40 and 250 atoms and so narrow that the atoms can move only along it: a one-dimensional gas, with a harmonic trap along the tube. A short sequence of light pulses set each atom into a quantum superposition of moving one way and the other, so that the gas in each tube split into two clouds with opposite momenta. The two clouds then oscillate in the trap, passing through each other at its centre twice per period (Figure 2).

Two clouds of gas colliding head-on thousands of times should thermalise: the momentum distribution of the atoms should relax to a single thermal hump. It did not. After thousands of collisions the gas had not noticeably equilibrated; the two populations were still there.

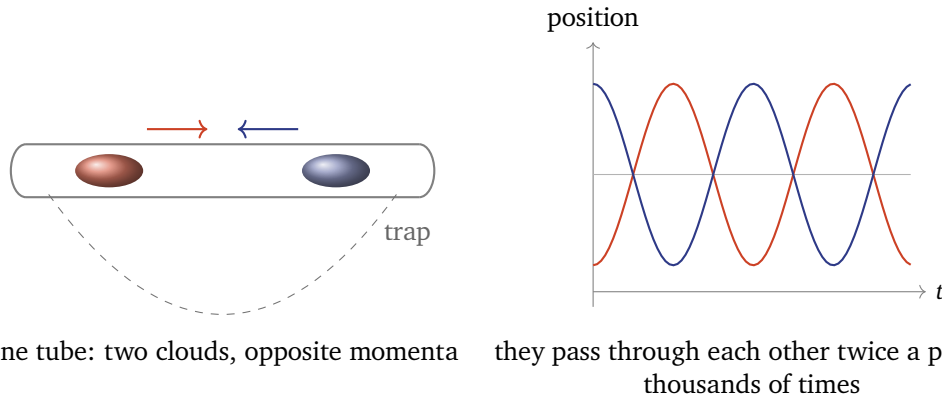


Figure 2: The quantum Newton's cradle of Kinoshita, Wenger and Weiss (2006), *schematically*: this is a cartoon of the set-up, not their data. Atoms confined to move along a tube, in a harmonic trap, are split into two clouds with opposite momenta; the clouds oscillate and collide at the centre twice per period.

The name is the desk toy's: a row of steel balls in which the momentum given to one end passes down the row and leaves at the other, over and over, without ever spreading out.

What the two have in common

Both experiments watched a system of many interacting parts and waited for it to forget its starting point. Neither did. In statistical mechanics the only things a system is allowed to remember are its conserved quantities, and what thermal equilibrium means is that the energy (with total momentum and particle number, where they are conserved) is *all* it remembers. A system that remembers more, for as long as these did, is evidence of further constraints on the motion — exact conserved quantities, or approximate ones that hold for a long time. *Integrability* is the case in which there are enough exact ones to pin the motion down; the FPUT chain is close to such a case, and the cradle's gas is very close (§ 0.3). This course makes that precise twice, classically in Part I and quantum mechanically in Part II, and then comes back to both experiments: FPUT in § 0.3 and Lecture 9, the cradle in Lecture 23.

0.2 Why one dimension is special

The Newton's cradle already contains the key fact, and it takes two lines of algebra.

Proposition 0.1 (Two equal masses on a line). *Two particles of equal mass collide elastically on a line, with velocities v_1, v_2 before and w_1, w_2 after. Then either $(w_1, w_2) = (v_1, v_2)$ or $(w_1, w_2) = (v_2, v_1)$: the particles pass unchanged or exchange velocities.*

Proof. Momentum and energy are conserved (elastic means the energy of the relative motion is returned in full once the particles separate):

$$w_1 + w_2 = v_1 + v_2, \quad w_1^2 + w_2^2 = v_1^2 + v_2^2.$$

Squaring the first and subtracting the second gives $w_1 w_2 = v_1 v_2$. So w_1 and w_2 have the same sum and the same product as v_1 and v_2 : they are the two roots of the same quadratic $x^2 - (v_1 + v_2)x + v_1 v_2 = (x - v_1)(x - v_2)$. ■

For particles that cannot pass through each other, only the exchange is possible. The conclusion needs equal masses. For masses $m_1 \neq m_2$ the same two laws give $w_1 = ((m_1 - m_2)v_1 + 2m_2v_2)/(m_1 + m_2)$, and $w_1 - v_2 = (m_1 - m_2)(v_1 - v_2)/(m_1 + m_2)$ is not zero: the velocities are reshuffled into new values. It also needs one dimension: in the plane, three conservation laws (two components of momentum, and energy) for four unknown velocity components leave a one-parameter family of outcomes, labelled by the scattering angle, and that is how a gas in two or three dimensions shares out energy.

So in one dimension, a gas of identical particles cannot change its *set* of velocities by two-body collisions at all. The list of velocities is merely handed from particle to particle. If the velocity distribution is to relax, something else has to do it: three particles close together at once.

Three bodies

For three equal masses the count changes. Momentum and energy are still two equations, but now for three unknown outgoing momenta, and two equations in three unknowns leave a one-parameter family. The family is a circle: with P the total momentum,

$$p_1^2 + p_2^2 + p_3^2 = \frac{P^2}{3} + u^2 + v^2, \quad u = \frac{p_1 - p_2}{\sqrt{2}}, \quad v = \frac{p_1 + p_2 - 2p_3}{\sqrt{6}},$$

so fixed P and fixed energy E confine the outgoing momenta to the circle $u^2 + v^2 = 2E - P^2/3$ (scripts/check-10-algebra.py, group 3). The six rearrangements of the incoming momenta are six points on it. A generic three-body collision lands somewhere else on the circle, and the set of momenta changes. That is how a one-dimensional gas can thermalise.

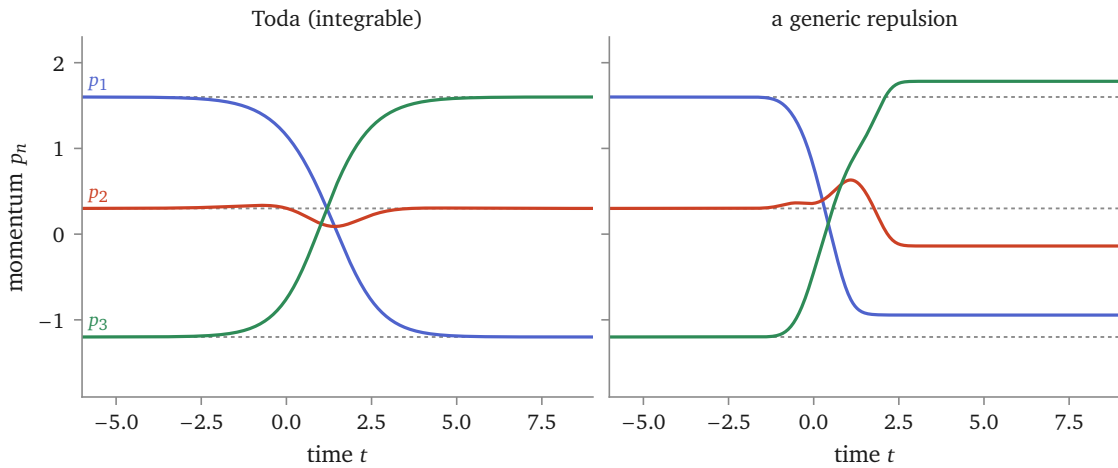


Figure 3: Three equal masses meet at once on a line, with the same incoming momenta (1.6, 0.3, -1.2) (dashed) in both panels. Left: the Toda interaction $e^{q_1 - q_2} + e^{q_2 - q_3}$ of Lecture 9. The outgoing momenta are the incoming ones, rearranged: p_1 leaves with -1.2, p_3 with 1.6, p_2 with 0.3. Right: a generic repulsion, the same Gaussian bump $e^{-(q_i - q_j)^2}$ between every pair. Momentum and energy are conserved in both runs, but on the right the particles leave with (-0.94, -0.14, 1.78), a new set, 0.44 away from every rearrangement (scripts/figures/collisions_1d.py).

Figure 3 runs the experiment twice. On the right, a reasonable short-range repulsion; the three particles leave with new momenta. On the left, the Toda chain, an interaction chosen

so that it has as many conserved quantities as particles — and the three particles leave with the momenta they came in with, rearranged, exactly as if they had met in pairs. [Lecture 9](#) explains why. The chain carries a matrix, its Lax matrix, which changes in time but only in a way that keeps its eigenvalues fixed; long before and long after the collision it becomes diagonal, with the momenta on the diagonal. So the incoming and the outgoing momenta are both its eigenvalues (the separation of the particles, which this uses, is a cited theorem there). For such a system, in scattering, the rest of the argument goes through: no collision, of two particles or of three or of N , changes the set of momenta the particles carry away. A gas is harder, because its particles are never free for long and their momenta at any instant do change. What an integrable gas conserves instead are quantities that play the part of the asymptotic momenta (in Part II, the *rapidities*), and a conserved distribution of those is enough to stop the gas relaxing to ordinary thermal equilibrium; [Lecture 16](#) and [Lecture 23](#) make this precise. The atoms of the Newton's cradle interact, to a good approximation, by point-like contact forces, and the one-dimensional Bose gas with point-like interactions is exactly such a system in its quantum form — the Lieb–Liniger gas of [Lecture 16](#). Kinoshita, Wenger and Weiss themselves suggested this as the probable explanation. The trap along the tube, and the tube's finite width, spoil it slightly, and why the experiment still did not thermalise on its time scale is [Lecture 23's](#) business.

0.3 What integrability means, twice

Two pictures have now appeared, one of energy returning to a mode and one of momenta being rearranged. They are the two faces of integrability, and the course treats both.

Classically: enough conserved quantities to straighten the motion

A system with n degrees of freedom — n coordinates, n momenta — always has the energy as one conserved quantity. It is *integrable* when it has n of them, independent and compatible with one another in a sense [Lectures 1](#) and [4](#) make precise. The Liouville–Arnold theorem ([Theorem 4.8](#)) says what that buys: under hypotheses [Lecture 4](#) states exactly — among them that the motion stays in a bounded region — a trajectory is confined to an n -dimensional torus, and in suitable angle coordinates it winds around that torus at constant speed. A torus is the surface of a doughnut when $n = 2$; a winding at constant speed is a straight line on it. The motion is *quasi-periodic*: a superposition of n periodic motions, like the normal modes of the Hooke chain, but with the modes bent into curved coordinates.

The difference is visible even with $n = 2$. Take the three-particle periodic Toda chain, which is integrable ([Lecture 10](#)), and remove its centre of mass: a point moving in a plane, (x, y) , in a potential well. Its cubic Taylor truncation is the Hénon–Heiles potential of 1964, a standard model of a star moving in a galaxy, which is not integrable. Follow one trajectory of each, from the same starting point at the same energy, and put down a dot at (y, p_y) — height and vertical momentum — every time the moving point crosses the vertical line $x = 0$ going right.

[Figure 4](#) is the result, and the reason for it is a count. At a crossing, $x = 0$, and the horizontal momentum p_x is fixed by the energy once y and p_y are known. A second conserved quantity K then gives one more equation, $K(0, y, p_x, p_y) = K_0$, on the two numbers (y, p_y) : in general the dots must lie on a curve. That curve is the slice of the trajectory's torus by the crossing line. Hénon–Heiles has no second conserved quantity of this global kind ([Lecture 6](#) states exactly which kinds are ruled out, and the islands of regular motion in its sections

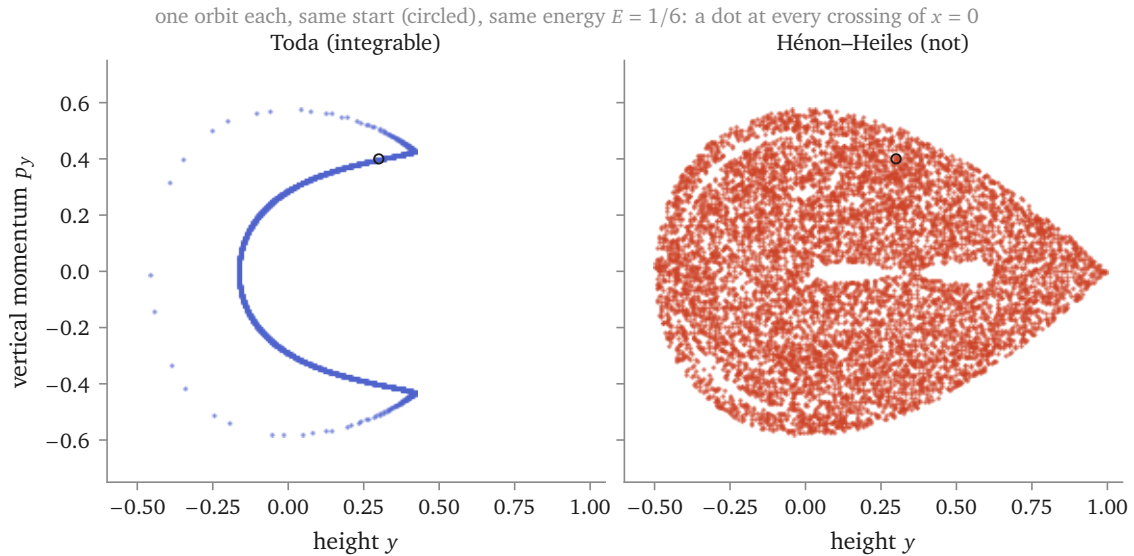


Figure 4: One trajectory each of the three-particle Toda chain (left) and of its cubic truncation, Hénon–Heiles (right), started from the same point (circled) at the same energy $E = 1/6$. A dot is drawn at every crossing of the line $x = 0$: 8000 of them in each panel. The Toda dots lie on a curve (dense where it is crossed slowly, sparse elsewhere); the Hénon–Heiles dots fill an area, apart from a few holes that belong to other trajectories. The generator measures the box-counting dimension of each cloud of dots, 0.8 and 1.9 (`scripts/figures/toda_hh_orbits.py`).

show that order survives on parts of phase space). This orbit is not in an island, and its dots spread over an area. Note what does *not* tell the two apart: the path in the (x, y) plane itself. Followed long enough, this Toda trajectory passes through every cell of a fine grid laid over the inside of the region its energy allows, just as the Hénon–Heiles one does. The constraint is on how position and momentum go together, and it shows up only when both are recorded.

Lecture 6 makes this construction, the *Poincaré section*, into a diagnostic, and shows that the two potentials, which agree through cubic order, disagree about everything that matters (Figure 6.9). The FPUT chain fits the same pattern. Its spring (0.2) agrees with a Toda spring through cubic order, for every $\alpha \neq 0$ (the tutorial at the end of this lecture), so at low energy the FPUT chain is a small perturbation of a Toda chain with the same fixed ends — which is integrable too, being the mirror-symmetric motions of a periodic Toda chain of $2N + 2$ particles (Toda, §2.10) · *cited*. Numerical studies find that at low energy, and for very long times, its motion stays close to the Toda chain’s (Benettin, Christodoulidi and Ponno 2013) · *cited*. On that reading, the recurrence is the shadow of the Toda chain’s conserved quantities. The FPUT chain itself is *not* integrable, and this course does not prove the statement about it, nor the fixed-end integrability just cited. What it proves is the Toda chain with the two other boundary conditions: Lecture 9 builds the open chain from the FPUT spring and proves its conserved quantities, and Lecture 10 finds the tori of the periodic one.

Quantum mechanically: many-body scattering is two-body scattering

In quantum mechanics “as many conserved quantities as degrees of freedom” turns out to be a poor definition: a Hamiltonian on a finite-dimensional space — a spin chain, say — commutes with the projector onto each of its own eigenstates, so conserved quantities are never in short supply, and what “independent” should mean has no agreed answer (Caux and Mossel 2011; [Lecture 13](#)). The working definition is the one [Figure 3](#) already showed. In an integrable system of particles on a line,

- the *set* of momenta is conserved: particles scatter off one another but only rearrange their momenta, never produce new ones; and
- scattering of N particles *factorises*: it is a sequence of two-body events, and everything about it — in quantum mechanics, the scattering amplitude — is built from the two-body data alone.

The many-body problem is then determined by the two-body problem. The Toda chain does this classically ([Lecture 9](#)); the quantum gases and spin chains of [Lecture 15](#) and [Lecture 16](#) do it quantum mechanically, and it is what the Bethe ansatz, the method of Part II, is built on.

The two faces are not equivalent definitions, and no theorem in this course turns one into the other. They are complementary diagnostics, and in the models studied here both apply. Particles confined to a ring cannot scatter off to infinity, and there the classical face is the relevant one: once its centre of mass is set aside, the periodic Toda chain of [Lecture 10](#) moves on tori. Its relative on an open line, the open chain of [Lecture 9](#), scatters and shows the second face. The course needs both.

0.4 The one miracle

Suppose the second face holds: three-body scattering is a sequence of two-body events. There are two ways for three lines to cross pairwise. Draw each particle’s path in space and time (a *world line*), with time running up. Three world lines with three different speeds meet pairwise in three crossings, and the middle line can pass either below or above the crossing of the other two ([Figure 5](#)). The two orders are different sequences of two-body events, and they describe the same three-body collision. They must agree.

This requirement is the *Yang–Baxter equation*, and it is a condition on the two-body scattering alone. For particles with nothing but a momentum it is automatic: each two-body event then contributes a number (classically, a shift in position; quantum mechanically, a phase), and numbers do not care about order. It bites when the particles carry an internal label — a spin, a colour — that a two-body event can exchange. Then each event is a matrix, matrices do not commute, and the equation is a strong constraint: a system whose two-body scattering fails it cannot scatter factorisably. Yang met it in 1967 as the consistency condition for exactly this kind of many-body scattering; Baxter met it in 1972 in the statistical mechanics of two-dimensional lattices. It has a classical shadow, the classical Yang–Baxter equation of [Lecture 8](#), whose picture ([Figure 8.4](#)) is this one; the Toda chain’s version is [Figure 9.6](#). Its quantum form is derived from three-body consistency in [Lecture 16](#). From [Lecture 17](#) on it is the consistency condition behind the algebraic method of Part II: a solution of it, together with a compatible model to attach it to, yields a family of commuting conserved quantities ([Lecture 17](#)), and when the model also has a suitable reference state, the algebraic Bethe ansatz builds eigenvectors, their eigenvalues and the Bethe equations

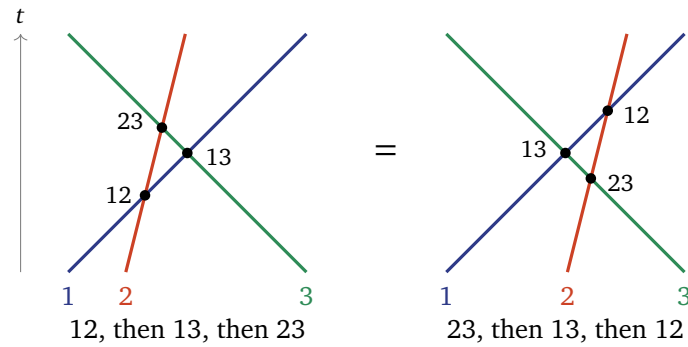


Figure 5: The picture behind the Yang–Baxter equation. Three particles, the same three speeds in both panels, time running up; only the starting offsets differ. The middle world line passes on one side of the crossing of the other two or on the other, so the three two-body events (dots, labelled by the pair) happen in opposite orders. If three-body scattering is nothing but its two-body events, both orders must give the same result. This is the picture the course keeps returning to: [Figure 8.4](#) classically, [Figure 9.6](#) for the Toda chain, and [Lecture 16](#) and [Lecture 17](#) quantum mechanically.

([Lecture 18](#)); whether those eigenvectors are all of them is a separate question ([Lecture 15](#), cited). The reading at the end of this lecture writes the equation down for the simplest solution.

The equation does not organise everything. The geometry of Part I — phase space, the Liouville–Arnold theorem, action-angle variables, the failure of integrability — owes it nothing. Nor does all of the algebra: the Lax pairs of [Lecture 7](#) and the reduction of [Lecture 12](#) do not come from it. What rests on it is one construction: commuting transfer matrices built from an r -matrix. There the classical equation is the model-independent condition that makes the bracket consistent — sufficient, not forced, as [Lecture 8](#) explains — and it is realised by the periodic chain of [Lecture 10](#); the quantum equation plays the same part from [Lecture 17](#) on. (The open chain of [Lecture 9](#) has an r -matrix bracket too, but of another kind, and its involution does not need the equation.) That construction is the course’s spine from Part I into Part II.

0.5 The no-free-lunch diagram

Why study systems that are exceptional? Because there are two roads from nature to a prediction, and each charges a toll ([Figure 6](#)).

On the left, one writes down the model one believes and solves it approximately: perturbation theory about something simple, numerics, linearisation about a minimum. On the right, one replaces the model by an idealisation that can be solved exactly, and pays in the model instead. A truncation you can solve is worth as much as a truth you cannot, and it is worth more when it is the organising centre of the truth. The harmonic oscillator organises vibrational spectroscopy, and the hydrogen atom organises chemistry, in exactly this way: nobody believes a molecule is a set of oscillators, but every real spectrum is read as a perturbation of that one. Integrable models are the next such centres: exactly solvable, genuinely interacting, and the points from which nearby systems can be studied as perturbations. How far that works is a real question. [Lecture 6](#) answers it classically (the

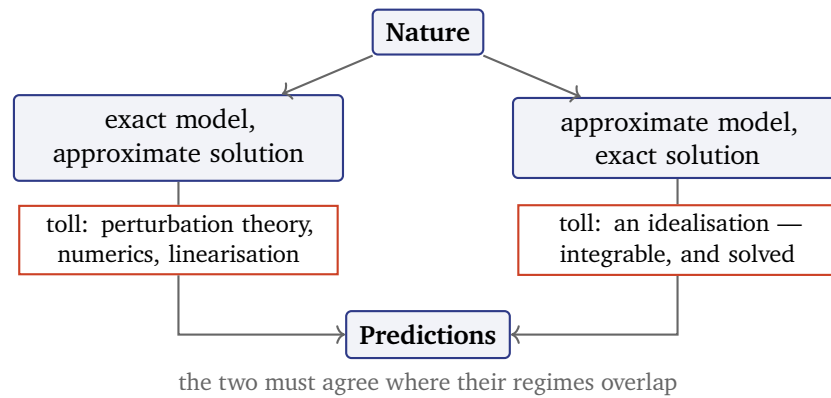


Figure 6: Two roads from nature to a prediction. On the left the model is exact and the solution approximate; on the right the model is an idealisation and the solution exact. Neither road is free.

KAM theorem, [Theorem 6.8](#): most of the tori survive a small perturbation, under hypotheses stated there), and [Lecture 23](#) says what happens to a nearly integrable quantum gas.

[Lecture 6](#) also adds the caveat that keeps this honest. Integrability is a property of the whole system, not of an approximation to it: truncating the integrable Toda potential at cubic order gives Hénon–Heiles, which is not integrable ([Figure 4](#)). The right-hand road does not mean “simplify until solvable”; it means finding the exceptional systems exactly as they are, and treating the rest of the world as their perturbation. Nor are the exceptional systems common: in a one-parameter family of potentials that contains Hénon–Heiles, the integrable members are isolated values of the parameter, for conserved quantities of a precise kind ([Lecture 6](#), which states exactly that much and no more).

0.6 The quadrant map

“Integrable” means different things in different places, and it helps to see all of them at once ([Figure 7](#)). Two questions sort them: is the system classical or quantum, and does it have finitely or infinitely many degrees of freedom?

The course lives in the left column. The lower left is the only quadrant in which the definition is settled and comes with a structure theorem, which is why Part I starts there and why it can be done with proofs. The upper left is where the definition is contested and the methods are powerful anyway; Part II takes the working criteria and makes each of them a lecture. The right column — solitons, the KdV equation that explained FPUT in the first place (the reading at the end of this lecture), integrable quantum field theory — is the sequel. It needs the left column first, and it appears here only as a signpost.

0.7 Map of the course

[Figure 8](#) is the course on one page. Part I builds the classical theory properly, in four stages: the geometry of phase space, integrability and its failure, the algebra that produces integrable systems, and the models on which all of it is tested. Part II quantises it one structure at a time.

The dashed arrows are the point. Most courses teach either classical integrability or quantum integrability. This one teaches the classical structures first, so that each quantum

	finitely many degrees of freedom	infinitely many (fields)
quantum	Quantum mechanics. No agreed definition. Working criteria: a transfer matrix built from a solution of Yang–Baxter; factorised scattering; Poissonian level statistics. Part II , Lectures 13–23.	Quantum field theory in two dimensions. Factorised S -matrix, form factors, thermodynamic Bethe ansatz. Not in this course.
classical	Classical mechanics. Settled: n independent compatible conserved quantities, and a theorem (Liouville–Arnold) that says what they buy. Part I , Lectures 1–12.	Classical field theory. Lax pairs and zero curvature, infinitely many conserved charges, solitons, the inverse scattering transform: KdV, sine-Gordon. Not in this course.

Figure 7: The quadrant map. Rows: classical or quantum; columns: finitely or infinitely many degrees of freedom. The course occupies the shaded left column. [Lecture 23](#) reprints this map with what the course covered marked on it.

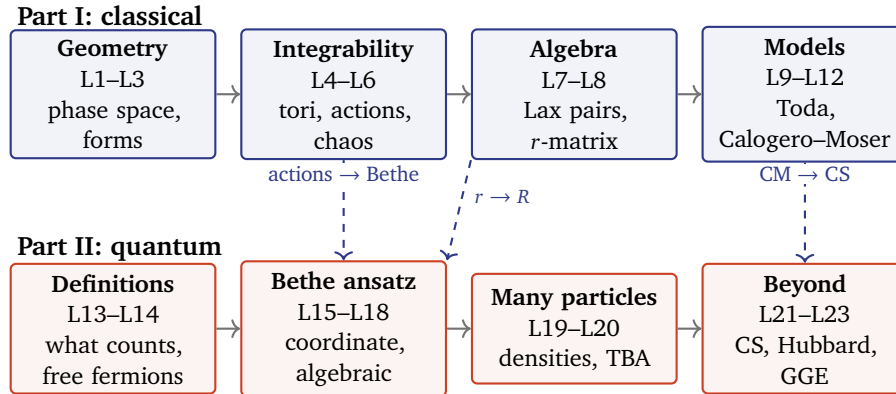


Figure 8: The course in eight blocks. Solid arrows: the order within each part. Dashed arrows: three places where a Part I structure is quantised into a Part II one — action variables into the Bethe equations, the classical r -matrix into the quantum R -matrix, the classical Calogero–Moser model into the quantum Calogero–Sutherland model.

object arrives as the deformation of something already owned:

Part I (classical)	Part II (quantum)
Poisson bracket $\{\cdot, \cdot\}$	commutator $[\cdot, \cdot]$
Lax matrix; monodromy and transfer matrices	the same names, now operators
classical r -matrix; classical Yang–Baxter	$R = \mathbb{1} + i\hbar r + O(\hbar^2)$; Yang–Baxter
action variables, $\oint p dq = 2\pi I$	the Bethe equations
momenta rearranged in scattering	factorised S -matrix

Every lecture of Part II carries a box naming its entry in this table.

What is assumed. Calculus, linear algebra (including eigenvalues and tensor products of matrices), ordinary differential equations, classical mechanics in Hamiltonian form, quantum mechanics at the level of spin and perturbation theory, branch points of multivalued functions (for [Lecture 10](#) only), and Bloch’s theorem. *What is not:* manifolds, differential

forms, symplectic geometry, Lie groups, Riemann surfaces, representation theory. Each is built, concretely, in the lecture that first needs it.

What is not done. Fields: the right-hand column of [Figure 7](#), with solitons, the KdV equation and the inverse scattering transform, is the sequel. And every result is marked: *proved* is the default and carries no mark, while *sketched* and *cited* are printed where they apply.

Reading

0.8 History, briefly

The FPUT computation was written up as a Los Alamos report in 1955 (Fermi, Pasta and Ulam, LA-1940). Mary Tsingou's part — she wrote and ran the programs — was brought to light by Dauxois (2008), and the problem now carries her name. The first explanation came from the continuum. Zabusky and Kruskal (1965) replaced the chain by a partial differential equation, the Korteweg–de Vries equation, found numerically that its localised waves pass through one another and re-emerge unchanged, called them *solitons*, and read the recurrence through their motion (Toda, §§1.1 and 2.7). That line leads to the inverse scattering transform and to classical field theory, the quadrant this course leaves out. Toda (1967) went the other way: keep the lattice, and look for a spring law for which the chain has exact nonlinear waves. He found the exponential spring of [Lecture 9](#), and the Toda chain turned out to be integrable. Whether the FPUT chain itself eventually equipartitions, and on what time scale, is still studied numerically, and the Toda chain is now the reference point for that question (Benettin, Christodoulidi and Ponno 2013).

Reading

0.9 The Yang–Baxter equation, written down

For readers who have met spin: let two particles each carry a spin- $\frac{1}{2}$, and let their two-body scattering act on the four-dimensional space $\mathbb{C}^2 \otimes \mathbb{C}^2$ of the pair by

$$R(u) = u \mathbb{1} + i P,$$

where P swaps the two spins and u is the difference of the two particles' rapidities, a parametrisation of their momenta. For three particles, write $R_{12}(u)$ for $R(u)$ acting on the spins of particles 1 and 2 and as the identity on the third, and so on. The two orders of [Figure 5](#) are the two sides of

$$R_{12}(u - v) R_{13}(u) R_{23}(v) = R_{23}(v) R_{13}(u) R_{12}(u - v),$$

an identity between 8×8 matrices. For this R it holds for all u and v ; for the apparently harmless variant $u \mathbb{1} + i(1 + cu)P$ it fails for every $c \neq 0$ (`scripts/check-10-algebra.py`, group 6). This is Yang's solution (1967), the one behind the Heisenberg chain of [Lecture 15](#). [Lecture 16](#) derives the equation from three-body consistency, and [Lecture 17](#) builds the rest of Part II on it.

0.10 The FPUT spring is a Toda spring, to cubic order

The general Toda spring, with two constants a, b , is

$$V_T(r) = \frac{a}{b} e^{-br} + ar - \frac{a}{b} = \frac{ab}{2} r^2 - \frac{ab^2}{6} r^3 + O(r^4),$$

the constant chosen so that $V_T(0) = 0$; the linear terms cancel. Match it to $\kappa(\frac{1}{2}r^2 + \frac{\alpha}{3}r^3)$, the FPUT spring (0.2) with a spring constant $\kappa > 0$. The quadratic terms give $ab = \kappa$ and the cubic terms $-ab^2/6 = \kappa\alpha/3$, so

$$b = -2\alpha, \quad a = -\frac{\kappa}{2\alpha}.$$

For every $\alpha \neq 0$ there is exactly one Toda spring that agrees with FPUT's through cubic order. For $\alpha < 0$ both constants are positive; with $\kappa = 1$, $\alpha = -\frac{1}{2}$ it is $e^{-r} + r - 1$, the spring of [Lecture 9](#). For $\alpha > 0$, as in [Figure 1](#), both are negative, and $V_T(r)$ is the $\alpha < 0$ spring with r replaced by $-r$: the chain of [Figure 1](#) is, to cubic order, a Toda chain with compression and extension exchanged (`scripts/check-10-algebra.py`, group 4). The two springs part company at quartic order, and that difference is what makes FPUT a perturbation of Toda rather than Toda itself.

Summary.

- *Two experiments* — the FPUT chain's recurrence and the quantum Newton's cradle — show many-body systems that do not forget how they were started ([Figures 1](#) and [2](#)): evidence of constraints beyond the energy, exact or approximate. Integrability is the case of enough exact ones.
- *One dimension is special*: two equal masses on a line can only exchange velocities ([Proposition 0.1](#)). A three-body collision has a circle of possible outcomes; an integrable one picks a rearrangement ([Figure 3](#)).
- *Integrability, classically*: n compatible conserved quantities confine bounded motion to tori; one orbit's crossing dots lie on a curve ([Figure 4](#)). Made precise in [Lecture 4](#).
- *Integrability, in scattering*: the set of momenta is conserved and many-body scattering is a sequence of two-body events. For Toda in [Lecture 9](#); the basis of the Bethe ansatz in Part II.
- *The one miracle*: the two orders of a triple crossing must agree — the Yang–Baxter equation ([Figure 5](#)), the consistency condition behind the commuting transfer matrices of [Lecture 10](#) and [Lecture 17](#).
- *Two roads, four quadrants*: an exactly solved idealisation is a road to prediction with its own toll ([Figure 6](#)); this course is the left column of [Figure 7](#).

References

M. Toda, *Theory of Nonlinear Lattices*, §1.1 (the FPU problem, its parameters and the recurrence) and §2.7; E. Fermi, J. Pasta and S. Ulam, Los Alamos report LA-1940 (1955); T. Dauxois, *Physics Today* 61(1) (2008) 55; N. J. Zabusky and M. D. Kruskal, *Phys. Rev. Lett.*

15 (1965) 240; M. Toda, J. Phys. Soc. Japan 22 (1967) 431; G. Benettin, H. Christodoulidi and A. Ponno, J. Stat. Phys. 152 (2013) 195; T. Kinoshita, T. Wenger and D. S. Weiss, Nature 440 (2006) 900; J.-S. Caux, B. Doyon, J. Dubail, R. Konik and T. Yoshimura, SciPost Phys. 6 (2019) 070 (the cradle by generalised hydrodynamics, [Lecture 23](#)); M. Hénon and C. Heiles, Astron. J. 69 (1964) 73; J.-S. Caux and J. Mossel, J. Stat. Mech. (2011) P02023; C. N. Yang, Phys. Rev. Lett. 19 (1967) 1312; R. J. Baxter, Ann. Phys. 70 (1972) 193, and *Exactly Solved Models in Statistical Mechanics*.